



Analytics Solutions Bootcamp: #01

Foundations of Analytics Solutions

Learn what analytics solutions are and how they solve business problems.

Hello, everyone! Welcome to the first module of the Analytics Solutions Bootcamp. In this module, you will learn about the Foundations of Analytics Solutions. We will talk about what it is and how we design and build the analytics solutions that will address our business problems.

ADD DATA PRODUCTS

Topics to be covered



- What is an analytics solution?
- Analytics solutions across data practices
- Real-world analytics solutions examples across industries
- Approach to designing and building analytics solutions
 - INFORMS Analytics Framework
- Introduction to data careers and career progression
 - AAP Philippine Skills Framework

What is an analytics solution?

Definition of analytics solution >>

Analytics is the process of collecting and analyzing data to discover, interpret, and communicate meaningful insights.

A **solution** is a combination of products, services, processes, and technologies designed to address a specific customer problem or meet a strategic business need.

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Let's define **analytics solution** word-by-word. [CLICK].

Analytics is the process of collecting and analyzing data to discover, interpret, and communicate meaningful insights. Every time we hear the term analytics, let's remember to think of the entire analytics process: from data collection, ingestion, cleansing, analysis, visualization, modeling, and finally presentation.

And the next work that we will define is solution. [CLICK].

A solution addresses a business problem or a need. It can be in a form of a product, service, creation or revision of an existing process, or technology. Every time you hear the term solution, always ask yourself "what problems are we trying to solve with this solution?".

So those are the definitions of the two words. Let's now put them together. [CLICK].

What is an analytics solution?

Analytics Solution in a nutshell >>

Analytics Solution aims to address business problems through data analytics.

1. We have a **clear** business problem.
2. We have a **clear baseline measure** of how we can quantify the problem.
3. We have **data** to analyze and explore.

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Analytics solution comes into play when we use data and our analysis skills to address a business problem. Later we'll enumerate some of the common real-world problems across different industries and what analytics solutions that we can make to address these problems.

Let's run through the high-level requirements when making an analytics solution. [CLICK].

- We have a **CLEAR** business problem. Emphasis on **CLEAR**. When we are creating a solution, sometimes, there are several back-and-forth. Example, you have a dashboard. You endorsed it to the users, and they returned it with some feedback and additions. If you're new to the job, you'll probably do it. You endorsed it again, but they returned with a new set of adjustments. Vice versa. When that happens, it's usually because the requirements are **NOT CLEAR OR INCOMPLETE**. [ASK] What do you guys think? Who is at fault when the requirements are lacking or incomplete? [CHECK ANSWERS]. [POSSIBLE EXPLAIN]. The users need to know what they really want, but it's your job to document the requirements. [CLICK].
- We have a clear **BASELINE MEASURE** of how we can quantify the problem. Usually, you will get requests on "making the reports more efficient". The users

need to quantify what we meant by “more efficient”. Most of the time, it means that they have reports that takes 2 days to do and want to automate the process. Whatever the problem that they presented, ensure that there is a clear measure on how they quantified the problem.

- We have **DATA** to analyze and explore. Sometimes, people say that they have data. But that doesn’t mean that they have a **WORKABLE DATA**. With that, you need to scrutinize their data sources. Identify all challenges that they have on their data and process. Yes, you might introduce changes in their process to improve it.

Types of analytics solutions

Analytics Solutions Examples >>

Domain	Solution
Data Engineering	Data ingestion pipelines
	Database migration projects
	Data warehouse / data lake implementation
Data Analysis / Business Intelligence	Executive reports
	KPI dashboards
	Financial reports

Domain	Solution
Data Science	Forecasting model
	Classification model
	Recommendation engine
Data Governance	Customer Unification (C360)
	Master Data Management
	Data cataloging

Real-world analytics solutions examples across industries

Real-world industry examples >>

Industry	Business Problem	Analytics Solution
Human Resources	Limited visibility into workforce composition and trends	Employee demographics dashboard
	High employee turnover	Employee attrition analysis
Retail	Understanding customer behavior	Customer segmentation
	Forecasting demand	Sales forecasting model
Banking / FinTech	Customers dropping out of account opening process	Funnel analysis
	High customer churn	Customer churn analysis
	Fraudulent transactions	Fraud detection and alerting solution

Real-world analytics solutions examples across industries

Real-world industry examples >>

Industry	Business Problem	Analytics Solution
Real Estate	Difficulty tracking financial performance	Financial reporting dashboard (AR, budget vs actual, cash flow)
	Low occupancy rates	Occupancy and tenant analytics dashboard
	Uncertainty in property pricing	Property valuation model
E-Commerce	Low conversion rates	Website funnel analysis
	Poor product recommendations	Recommendation engine
	High cart abandonment	Cart abandonment analysis
Healthcare	Long patient waiting times	Patient flow analytics
	Resource allocation challenges	Hospital utilization dashboard

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Real-world analytics solutions examples across industries

Real-world industry examples >>

Industry	Business Problem	Analytics Solution
Manufacturing	Unexpected equipment failures	Predictive maintenance solution
	Production bottlenecks	Production efficiency dashboard
	Product quality issues	Quality control analytics
Logistics & Supply Chain	Late deliveries	Delivery performance dashboard
	Inefficient routes	Route optimization analytics
	Inventory imbalances	Supply chain visibility dashboard
Telecommunications	Customer attrition	Subscriber churn prediction
	Network disruptions	Network monitoring analytics
	Service quality concerns	Customer experience dashboard

Real-world analytics solutions examples across industries

Real-world industry examples >>

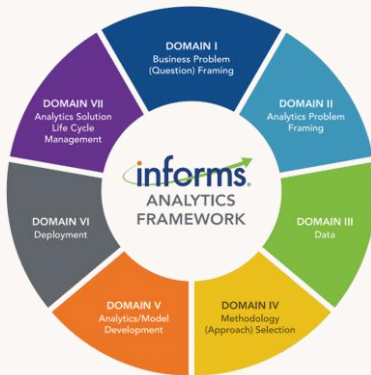
Industry	Business Problem	Analytics Solution
Education	Student dropouts	Student retention analysis
	Low academic performance	Academic performance dashboard
	Resource planning challenges	Enrollment forecasting
Marketing	Ineffective campaigns	Marketing campaign performance dashboard
	Poor lead conversion	Lead funnel analytics
	Unclear return on investment	Marketing ROI analysis
Government / Public Sector	Slow service delivery	Public service performance dashboard
	Resource allocation challenges	Budget utilization analytics
	Lack of citizen insights	Citizen feedback analysis

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INFORMS Analytics Framework >>



1. Business Problem Framing
2. Analytics Problem Framing
3. Data
4. Methodology Selection
5. Analytics/Model Development
6. Deployment
7. Analytics Solution Lifecycle Management

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DOMAIN I: BUSINESS PROBLEM (QUESTION) FRAMING Analytics methodologies are applied to problems, which are sometimes stated as questions. The Business Problem (Question) Framing Domain includes activities related to understanding the business problem and relevant questions, as well as evaluating the scope of the problem. The activity may also include the identification of opportunities, challenges, or just an evaluation if a question is well defined. This framing process is essential for ensuring that a clear, actionable problem is defined before seeking solutions. Tasks in this Domain include determining, refining, and clarifying a business problem statement; identifying stakeholders and aligning the stakeholders with the stated business problem statement; and determining whether the problem is amenable to an analytics solution. With agreement on a business problem statement and appropriate sponsor agreement to move forward with a solution, the remaining Domains of the INFORMS Analytics Framework are now applicable.

DOMAIN II: ANALYTICS PROBLEM FRAMING The Analytics Problem Framing Domain focuses on translating a clearly defined business problem (question) into a structured analytics problem that can be addressed with analytics techniques applied to data. This process ensures that analytical efforts are

aligned with business objectives and that outcomes will be relevant and actionable. A significant activity in this Domain is reformulating the business question into an analytics problem, requiring the application of descriptive, predictive, or prescriptive analytics. This translation helps define the type of analytics or model needed. Another activity is identifying the critical drivers or inputs and determining how they relate to desired outputs, forming the basis for hypothesis testing or model design. To ensure relevance and effectiveness, the Domain includes defining primary success measures, such as accuracy, revenue lift, or operational efficiency gains. Risk identification and mitigation strategies are also important to anticipate challenges such as data limitations or unintended bias. This Domain emphasizes the need for sponsor agreement and stakeholder alignment to ensure buy-in and shared understanding before moving forward with analytics solution development. By clearly framing the analytics problem, organizations can reduce ambiguity, align technical efforts with strategic goals, and set a solid foundation for model development and implementation

DOMAIN III: DATA The Data Domain encompasses activities related to identifying the data required and available, ensuring its usability for solving analytics problems; and fulfilling documentation and reporting needs. It also includes the effective utilization of data, maintaining its privacy and security, ensuring proper data transport across systems, and adhering to best practices for data governance. Making data usable includes critical actions applied to the data before it can be used, such as cleaning, harmonizing, transforming, merging/joining and validating data, as well as data quality evaluation. As a critical asset, security, management, and privacy protocols are vital to support business needs to ensure compliance, ethical standards, and trust in output and outcomes from derived data. This Domain also addresses the elements of what data is owned, used, and produced. Understanding the use of data inventory and documentation is incorporated in this Domain to ensure repeatable processes and the ability to manage the volume, velocity, variety, and veracity of data.

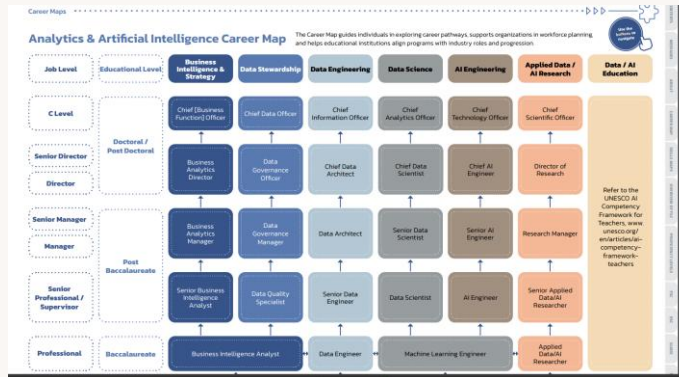
DOMAIN IV: METHODOLOGY (APPROACH) SELECTION The Methodology (Approach) Selection Domain involves selecting the appropriate analytical methods, tools, and technologies to solve the defined analytics problem. This ensures that the chosen approach aligns with the problem type, available data, and resource constraints. The Domain includes identifying suitable analytical techniques based on the nature of the problem, as well as evaluating and selecting the best-fit methods considering available data, timeframes, and organizational resources. The Domain includes the selection of a compatible technology stack to effectively implement the solution.

DOMAIN V: ANALYTICS/MODEL DEVELOPMENT The Analytics/Model Development Domain covers activities related to designing, building, and refining models, as well as documenting and communicating the results. The Domain incorporates designing the model structure, building the appropriate models, and running and evaluating model performance to ensure reliability. It also addresses identifying, using, calibrating, and integrating models as well as the documentation of the model and communication of findings. The Domain emphasizes documenting and communicating findings, including assumptions and limitations, to maintain transparency. This Domain ensures that models are reliable, well-understood, appropriately tested, and effectively support decision-making.

DOMAIN VI: DEPLOYMENT The Deployment Domain covers activities related to delivering the analytics solution and ensuring it is effectively integrated into the business environment. This involves performing business validation of the solution and a report of the validation findings. It includes securing sponsor agreement and stakeholder alignment to proceed with deployment. The deployment activities support the implementation, migration, and testing of the solution for production, ensuring it functions as intended. Analytics plays a key role in this Domain, as it involves validation and verification, working in conjunction with technology management, and production data flows. This Domain also includes the role of analytics in evaluation of deployment challenges to determine best resolution through data, model, technology, or operational environment conditions. This Domain ensures the analytics solution is successfully adopted, integrated, and delivers business value.

DOMAIN VII: ANALYTICS SOLUTION LIFECYCLE MANAGEMENT The Analytics Solution Lifecycle Management Domain involves ongoing oversight and calibration to ensure the analytics solution continues to perform effectively and delivers valid results over time that continue to deliver business value. Ongoing training activities are incorporated in this Domain to ensure optimal performance management. The Domain includes tracking solution performance and covering recalibration and maintenance to keep the solution accurate. Other tasks include periodically validating the business case to confirm continued value and ensuring documentation is kept up to date, taking into consideration context, environment, and the need for delivering effective end results. Continuous lifecycle management is critical in sustaining the impact and reliability of analytics solutions.

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